

A Fully Photometric Approach to Type Ia Supernova Cosmology in the LSST Era: Host Galaxy Redshifts and Supernova Classification

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The upcoming Vera C. Rubin Observatory's Legacy Survey of Space and Time (LSST) is expected to discover millions of Type Ia supernovae (SNe Ia), offering an unprecedented opportunity to constrain dark energy. The vast majority of these events will lack spectroscopic classification and redshifts, necessitating a fully photometric approach. We present detailed simulations based on the Extended LSST Astronomical Time Series Classification Challenge (ELAsTiCC), and a cosmological analysis using photometrically classified SNe Ia with host galaxy photometric redshifts. This dataset features realistic multi-band light curves, host mis-associations, and transient-host correlations across the high-redshift Deep Drilling Fields (DDF) ($\sim 50 \text{ deg}^2$), and we include a separate low-redshift sample based on the Wide Fast Deep (WFD) fields. We employ a joint SN+host photometric redshift fit, a neural network based photometric classifier (SCONE), and BEAMS with Bias Corrections (BBC) methodology to construct a bias-corrected Hubble diagram. We produce statistical + systematic covariance matrices, and perform cosmology fitting with a prior on Cosmic Microwave Background (CMB). We fit and present results for the w CDM dark energy model, and the more general Chevallier-Polarski-Linder (CPL) w_0w_a CDM model. Our results demonstrate that a fully photometric analysis provides a viable alternative to spectroscopic samples. With a simulated sample of ~ 5000 , we achieve Figure of Merit (FoM) of about 180. [Bias comment awaits further study]

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References

I. INTRODUCTION

The discovery of dark energy and the accelerating expansion of the Universe, first identified through the observation of a few dozen Type Ia supernovae (SNe Ia) [Riess et al. \(1998\)](#), [Perlmutter et al. \(1999\)](#), marked a pivotal moment in modern cosmology.

Subsequent SNIa surveys yielded constraints on the dark energy equation of state that agree well with a cosmological constant ($w = -1$) ([Riess et al. 2004](#), [Wood-Vasey et al. 2007](#), [Astier et al. 2006](#), [Conley et al. 2011](#), [Suzuki et al. 2012](#), [Betoule et al. 2014](#), [Scolnic et al. 2018a](#)), where w is the ratio of pressure to energy density for dark energy (for reviews see e.g. [Frieman et al. \(2008\)](#), [Weinberg et al. \(2013\)](#)). Pantheon+ ([Brout et al. 2022](#)) have largely agreed and further sharpened on the constraints on dark energy estimates. More recently, however, few- σ hints of evolving dark energy have emerged from analyses with higher statistics and better systematics handling from the Dark Energy Survey ([Abbott et al. 2024](#), DES-SN5YR), the Dark Energy Spectroscopic Instrument ([Adame et al. 2025](#), DESI), and the UNION sample Bayesian analysis ([Rubin et al. 2023](#)). While these deviations from a cosmological constant have not reached discovery significance, they have spurred considerable interest in the possibility that dark energy may be dynamic rather than static. In this context, upcoming projects such as the Rubin Legacy Survey of Space and Time ([Cahn 2009](#), [Ivezić et al. 2019](#), LSST) and the Roman Space Telescope ([Doré et al. 2019](#), [Zhai et al. 2020](#)) will detect tens of thousands of Type Ia supernovae to significantly reduce the statistical and systematic precision on parameters modelling the dark energy.

Historically, SN Ia cosmology surveys e.g., SNLS and SDSS-II at intermediate redshifts, and the CfA ([Riess et al. 1999](#), [Hicken et al. 2012](#)) and CSP ([Krisciunas et al. 2017](#)) programs at low- z , relied on intensive spectroscopic follow-up to confirm each transient's type and to secure a host-galaxy redshift. These spectroscopically confirmed samples collectively numbering to a few hundreds of SNe Ia set the early gold standard for supernova cosmology, forming the backbone of composite data sets like the Pantheon compilation of 1048 spectroscopic SNe Ia ([Scolnic et al. 2018b](#)) and its recent Pantheon+ extension with 1550 SNe Ia ([Brout et al. 2022](#), [Scolnic et al. 2022](#)).

Previous photometric surveys (SDSS-II, Pan-STARRS, DES-SN) acquired spectroscopic information for only a small fraction of their samples, and this

challenge will be even more pronounced in the LSST era, which will have a significantly higher detection rates. Therefore, for the vast majority of newly discovered SNe it has become necessary to develop photometric classification and a formalism to analyse a contaminated Hubble diagram.

Pan-STARRS1 ([Jones et al. 2018](#), [Jones et al. 2019](#)) used the PSNID classifier ([Sako et al. 2011](#)) and the BEAMS¹ formalism to determine cosmology constraints from a photometrically identified sample. For DES-SN5YR they used more advanced photometric classifiers based on neural networks: SuperNNova (SNN)² ([Möller and de Boissière 2020](#)) and SCONE ([Qu et al. 2021](#)) (see Sec. III C). Both are neural network based classifiers, and have exhibited very high levels of purity and accuracy $> 98\%$. They also used an extension of the BEAMS formalism, “BEAMS with Bias Correction” (BBC), that is described in [Kessler and Scolnic \(2017\)](#) [hereafter KS17].

While photometric classification has been successfully used to deliver cosmological results, photometric redshifts have not. A photometric redshift method using the SN+host galaxy photo- z has been proposed ([Kessler et al. 2010](#), [Palanque-Delabrouille et al. 2010](#), [Roberts et al. 2017](#)), and [Mitra et al. \(2023, M23\)](#) performed the first rigorous SNIa-cosmology analysis with photo- z 's and systematics, but did not include non-SNIa contamination.

As part of the DESC³ effort to prepare cosmology analyses for LSST data, here we simulate LSST data that includes both host galaxy photo- z 's and non-SNIa contamination, and we incorporate SN+host photo- z and BEAMS into the analysis. This analysis supersedes M23, and is applicable to real data. M23 presented a thorough light curves to cosmology analysis with photometric redshift (z_{phot}) using LSST simulated data. However, M23 was focused on evaluating systematics related to photo- z , and therefore the analysis is not practical on real data because of the simplifying assumption that every SNIa is perfectly classified (i.e. spectroscopic sample). A full-scale analysis involving both photometric classification and z_{phot} with extensive systematics has not yet been done. In this paper, in addition to using photometric redshift using host galaxies we also use photometrically identified SNIa. Here we include non-SNIa contamination in the sim and photometric classification in the analysis. Therefore we present the first realistic analysis (that can be applied on real data) on simulated LSST data, that combines both photometric classification and photometric redshifts (see table I). The historical trajectory and evolving methodologies of SNIa cosmology research are summarised in Table I.

This paper is organised as follows, in sec. II we describe the simulation process, necessary to generate the

¹ BEAMS = Bayesian Estimation Applied to Multiple Species

² <https://github.com/supernnova/SuperNNova>

³ Dark Energy Science Collaboration

Classification	Redshift	Survey/Project
Spectroscopic	Spec Only (SN or Host)	Two decades of results Hi-z (Riess et al. 1998), SCP (Perlmutter et al. 1999) SNLS (Astier et al. 2006), ESSENCE (Wood-Vasey et al. 2007), SDSS-II (Kessler et al. 2009), PS1 (Rest et al. 2014), Pantheon (Scolnic et al. 2018a,b), JLA (Betoule et al. 2014), DES-3YR (Abbott et al. 2019)
Photometric	Spec Only (SN or Host)	PS1 (Jones et al. 2018), DES-SN5YR (Vincenzi et al. 2024)
Spectroscopic	Spec or Photo (Host Only)	DES-5YR + redMaGiC (Chen et al. 2022), LSST DESC Sims [M23] (Mitra et al. 2023) (precursor analysis)
Photometric	Spec or Photo (Host Only)	Current Analysis

TABLE I: History of evolution of modern SNIa cosmology starting from spectroscopic redshift and classifications to photometric classifications and redshift.

data and for the bias corrections. In sec. III describe the main analysis of this paper. In sec. III A we present the light curve fitting process and the different selection cuts and systematic uncertainties considered. Then we present the photometric classification technique using SCONE in sec. III C and move to BBC analysis in sec. III D 2. The BBC stage produces the bias corrected distance estimates, which is then used to construct the Hubble diagram. This is then used for cosmology and dark energy estimation which is discussed in the next section in sec. IV. Under the cosmology section we present dark energy estimates for w -CDM cosmology (sec. IV A) and $w_0 w_a$ (CPL) cosmology (sec. IV B). In light of the current DES-SN5YR results, we present a comparison with our estimates, which is presented in sec. IV C. In sec. V A we discuss the motivation for using the current event selection strategy adapted in this analysis, and how it was different that that used in DES-SN5YR SNe selection strategy. Finally in sec. VI we present the conclusion of this analysis.

II. SIMULATED DATA

A. Overview

Ideally, we would work with simulated images from DC2 (Sánchez et al. 2020) or OpenUniverse2024 (OpenUniverse et al. 2025), and obtain light curves from the LSST difference imaging analysis (DIA)⁴ based on Alard and Lupton (1998). While such an analysis is most realistic, it is very CPU intensive and repeating the analysis is not practical. Instead, we simulate SN Ia light curves corresponding to the output of DIA.

We simulate and analyse 25 independent samples, where each sample includes two subsets: (1) a complete low- z ($z \sim 0.08$) sample of spectroscopically confirmed SNe Ia with accurate spectroscopic redshifts, and (2) a high- z ($0.03 - 1.55$) SN Ia sample with non-SNIa contamination from SNII/Ib/Ic and peculiar SNIax and 91bg-like. The high- z subset includes host-galaxy photo- z 's, but does not have spectroscopic redshifts nor identification.

The low- z cadence is from the the Wide Fast Deep (WFD) observing strategy from the Data Challenge 2 (DC2) (LSST Dark Energy Science Collaboration (LSST DESC) et al. 2021). The high- z cadence is from ~ 50 deg² Deep Drilling Fields (DDF)⁵. To model host properties and incorrect host associations, we use host galaxy libraries (HOSTLIBs) from the ELAsTiCC (Extended LSST Astronomical Time Series Classification Challenge). The simulation for SNIa and detection efficiency is the same in both subsets.

ELAsTiCC

To help prepare spectroscopic follow-up for transients and hosts found by Rubin, ELAsTiCC was developed to enable early and accurate photometric classification of transients using both light curves and contextual information (Foley and Mandel 2013, Baleschi et al. 2020, Gagliano et al. 2021). ELAsTiCC includes ~ 5 million detected transients and ~ 50 million alerts across 30+ classes, embedding realistic transient-host correlations from the CosmoDC2 catalog (Korytov et al. 2019). While we don't use the ELAsTiCC simulations here, we leverage the host simulation for ELAsTiCC that was developed to

⁴ https://github.com/LSSTDESC/dia_pipe

⁵ <https://www.lsst.org/scientists/survey-design/ddf>

model host-SN correlations, and we include a few improvements to model host mis-associations.

B. Implementation

The SNANA simulation includes three main steps as described in Kessler et al. (2019a): (a) estimate the true source magnitude from a rest-frame SED model that incorporates redshifting and cosmic expansion ($\Omega_M = 0.315$, $w = -1$, flatness), lensing magnification, peculiar velocity, Milky way extinction, and host galaxy extinction; (b) model the flux uncertainty and random fluctuations using sky noise, zero-point, and PSF for each observation; (c) model DIA detection efficiency vs. SNR.

We simulate SNe Ia using the SALT3 SED model (Kewenworth et al. 2021) with a NIR extension from Pierel et al. (2022) so that rest-frame i , z , Y wavelength ranges are included.

We use a dust based intrinsic scatter model, along with the stretch and color populations, from Brout and Scolnic (2021), Popovic et al. (2023). This model was developed for the DES-SN5YR analysis (Vincenzi et al. 2024). Type Ia supernovae in galaxies with higher masses ($x_M > 10$, where x_M is the log of stellar mass) are found to be ~ 0.05 mag brighter than those in less massive galaxies, even after correcting for light curve stretch and color (Sullivan et al. 2010, Lampeitl et al. 2010, Smith et al. 2020, Kelsey et al. 2021). Recent studies show this “mass-step” effect depends on SN color, with negligible impact on blue SNe but over 0.1 mag for redder ones (Brout and Scolnic 2021, Kelsey et al. 2023). This mass step has been attributed to dust, with Brout and Scolnic (2021) and Salim and Narayanan (2020) suggesting that varying R_V distributions as a function of stellar mass contribute the effect.

The stretch-luminosity parameter is $\alpha^{\text{true}} = 0.15$, and the intrinsic color-luminosity parameter $\beta_{\text{int}}^{\text{true}} = 2.12$. The combination of intrinsic color variation and dust results in an effective fitted $\beta \sim 3$ (sec 3.4 in Vincenzi et al. (2024)), but there is no true effective β value to compare with.

Correlations between SNe and host-galaxy mass are modelled using ELASTiCC HOSTLIBS and probability distributions.

The average 5σ limiting magnitudes per co-added night, and their average time between nights for each of the $ugrizY$ passbands, are listed in Table II for both the low- z (WFD) and the high- z (DDF) subsets. On average the DDF is ~ 1.0 mag deeper than the WFD, with g and r being the deepest bands. Compared to DDF, the WFD cadence has 30% fewer co-added observations on average. For host galaxies we use a 1 year co-added depth which corresponds to a depth of $i \sim 25.4$.

TABLE II: Average depth and time between observations.

Filter	WFD		DDF	
	depth ^a	gap ^b	depth	gap
u	23.84	10.5	25.05	5.3
g	24.80	11.9	25.52	7.3
r	24.21	8.2	25.60	7.3
i	23.57	8.6	25.19	7.3
z	22.65	9.0	24.79	7.3
Y	21.79	11.2	23.83	7.4

^a 5σ limiting magnitude.

^b Average time (days) between visits, excluding seasonal gaps.

Rates

We use the volumetric rate model $R(z)$ based on Dilday et al. (2008) for $z < 1$ and Hounsell et al. (2018) for $z > 1$. The rate $R(z)$ we adopt is given by

$$R(z) = 2.5 \times 10^{-5} (1+z)^{1.5} \text{ yr}^{-1} \text{ Mpc}^{-3} \quad (z < 1) \quad (1)$$

$$R(z) = 9.7 \times 10^{-5} (1+z)^{-0.5} \text{ yr}^{-1} \text{ Mpc}^{-3} \quad (z > 1) \quad (2)$$

The fraction of SNIa For the high- z sample, the number of generated events (N_{gen} columns of Table III) is computed from the measured volumetric rate, duration of the survey, and 50 deg^2 area of DDF. For low- z , N_{gen} is arbitrarily chosen such that the number of events after selection requirements is roughly about $\sim 10\%$ of the high- z statistics.

Peculiar SNe are simulated using Kessler et al. (2019b) and Vincenzi et al. (2021). While Core collapse SNe rates are simulated using templates from Vincenzi et al. (2019). Rates are calculated from Strolger et al. (2015) and Shivers et al. (2017), for more details see Vincenzi et al. (2024), Vincenzi et al. (2021).

For the low- z sample we arbitrarily generate 4200 sample, following the z dependence in eq.1. For the high- z sample we use the redshift range survey duration and sky area to compute the physical rate and equation (eq.1 and 2). High- z sample both Ia and non-Ia use a physical rate. The physical rates for 91bg type is $\sim 12\%$ of the Ia rate and has the same z dependence as in eq. (1). SNIax follows Madau and Dickinson (2014), following the SFR with $6e - 6/\text{yr}/\text{Mpc}^3$ at $z = 0$. For the core collapse events, we define R_{CC} to be the green curve in Fig. 6 in Strolger et al. (2015). The type II rate is $0.7 \times R_{CC}$, and the type Ibc rate is $0.3 \times R_{CC}$.

Host Galaxies

The HOSTLIBS for SNIa and non-Ia events are constructed as separate catalogues. To select the true host and possible host mismatch, all potential galaxies within a $10''$ radius are included for the d_{DLR} analysis described

in [IIIB 1](#). Roughly half the galaxies have no neighbours within $10''$, $\sim 1/3$ have a single neighbour and the remainder have 2 or more neighbours.

To describe the brightness profile of each galaxy, a single Sersic profile was estimated from [Gagliano et al. \(2021\)](#). The HOSTLIB Sersic parameters are assumed to represent the true galaxy profile, without instrumental smearing. A true SN location near a host galaxy is overlaid using this profile. To incorporate PSF effects we smear with an average FWHM $\sim 1.0''$. Thus there are two Sersic profiles: one for selecting SN location and the other for computation of d_{DLR} in the analysis.

Since each HOSTLIB is weighted based on the transient type the galaxy density is not accurate and therefore the neighbour fractions are not accurate. To have a host mismatch fraction that is similar to DES-SN5YR ([Qu et al. 2024](#)), we increase the Sersic size by a factor of three. To avoid this artifact in future simulations, an unweighted HOSTLIB should be used eg. the *diffsky*⁶ catalogue from *OpenUniverse2024*.

For the source of the photo- z redshift prior, we used photo- z PDF unpacked from quantiles (where accurate spec- z Gaussian prior is not available). The photo- z information is provided to the HOSTLIB files by passing 11 quantile redshift columns corresponding to an integrated cumulative density function's probabilities of 0, 10%...100%.

To characterise the photo- z residuals, we follow [Graham et al. \(2018\)](#) and define metrics for an inner core resolution and outlier fraction using the quantity $\Delta z_{(1+z)} = |z_{\text{phot}} - z_{\text{true}}|/(1 + z_{\text{phot}})$. The resolution is the RMS width of the inter quartile distribution of $\Delta z_{(1+z)}$, divided by 1.349, and is denoted by σ_{IQR} . The outlier fraction (f_{out}) is the fraction of events satisfying

$$\Delta z_{(1+z)} > 3\sigma_{\text{IQR}} \text{ and } \Delta z_{(1+z)} > 0.06 \quad . \quad (3)$$

For $z_{\text{true}} < 0.4$ nearly all of the transient events have a spectroscopic redshift, and for $z_{\text{true}} > 1.4$ the SNe are too faint for detection. For the relevant redshift range ($0.4 < z_{\text{true}} < 1.4$), $\sigma_{\text{IQR}} = 0.115$ and $f_{\text{out}} = 0.15$.

Weight Maps

Unlike the PLAsTiCC HOSTLIB used in [Mitra et al. \(2023\)](#), the use of ELAsTiCC HOSTLIBs enables modelling of the host galaxy correlations. Here we weight each transient host as a function of log of stellar mass (LOGMASS) and the star formation rate (SFR). The corresponding probability distribution is called a WGTMAP. They are tuned to Pan-STARRS (PS1) data ([Chambers et al. 2016](#)) (see [Gagliano et al. \(2021\)](#) for more details).

For each transient class, [fig. 1](#) shows the generated number of events as a function of SFR and LOGMASS plot for the host galaxies selected by SNANA. The green points show a representative sample from the ELAsTiCC HOSTLIB, and the yellow points show the host galaxies selected by the WGTMAP. For all transient classes, the WEIGHTMAPS preferentially select higher SFR, and for the 91bg WEIGHTMAP preferentially selects higher LOGMASS.

TABLE III: Summary of simulation statistics vs. sample and transient type.

Sample	Transient	Number of events		
		Total generated ^a	Passed trigger ^b	Passed cuts ^c
Low- z 0.01, 0.08	Ia	4200	1133	691 (0.165)
High- z 0.03, 1.55	All	233703	23479	5034 (0.022)
	Ia	35719	11393	4389 (0.13)
	Iax	18909	579	141 (0.008)
	91bg	4798	399	2 (~ 0)
	II	121994	8643	313 (0.003)
	Ibc	52283	2465	189 (0.004)
Final		5080 (Ia) + 645 (CC) = 5725		

^a Total number of generated SNe

^b Two or more detections separated by more than 30 minutes. The detections are not necessarily at peak brightness.

^c After selection cuts; number in '()' is fraction with respect to "Generated total" column.

Sample Sizes

A summary of average simulation statistics is shown in [Table III](#). For low- z we generate 4200 events, and $\sim 1/4$ satisfy the 2-detection trigger. For the high- z sample we generate nearly 36,000 SNIa, and $\sim 1/3$ satisfy the trigger. The high- z sample also includes nearly 200,000 non-SNIa, and $\sim 6\%$ satisfy the trigger. Compared the non-SNIa, the SNIa trigger efficiency is much higher because these events are significantly brighter.

To visualize the relative number of events in high- z sample [Fig. 2](#) shows several pie-charts corresponding to [Table III](#). Comparing physical rates, SNIa corresponds to 15% of all SNe ([Fig. 2a](#)), and after the trigger requirement, nearly 50% of the SNe are type Ia ([Fig. 2b](#)).

Examples of simulated light curves at different redshifts are shown in [Fig.1](#) of [Mitra et al. \(2023\)](#).

To train our photometric classifier ([Sec. III C](#)) and to implement distance bias corrections in BBC ([Sec. III D](#)), we perform a separate simulation to generate a large sample of millions of events. The simulation procedure is exactly identical to that used for the simulated data.

⁶ <https://diffsky.readthedocs.io/en/latest>

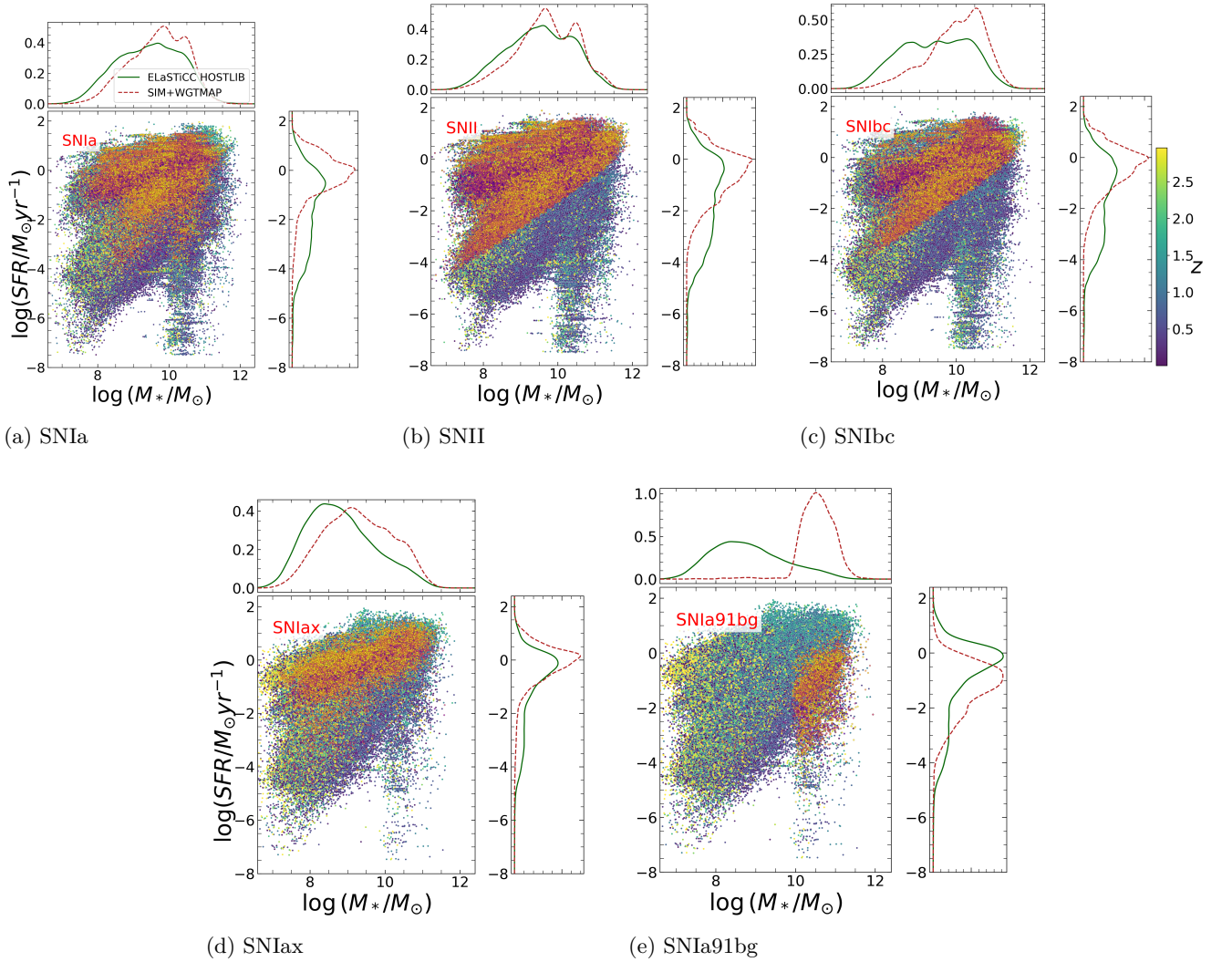


FIG. 1: Density of simulated galaxies as a function SFR vs LOGMASS. Each panel shows a different transient class. Figure illustrating how host galaxies of different type of SNe, exhibit different type of SFR and $M_{\odot}$ properties. The yellow (green) points correspond to ELAsTiCC HOSTLIB with (without) the WGTMAP. To avoid saturation we show random 10% of the HOSTLIB.

III. ANALYSIS

A. Lightcurve Fitting

The SNIa-cosmology analysis steps are shown in Fig. 3, and described below. This analysis is similar to the M23, with additional complexities, from using photometric classifier SCONE and BEAMS formalism.

While M23 performed separate analyses for z_{phot} and z_{spec} samples, here we focus our study on a single z_{phot} analysis with no spectroscopic redshift or spectroscopic classification. While we anticipate some spectroscopic resources for LSST-DDF, here we make the most conservative assumption of no resources in order to be more sensitive to potential problems in the analysis.

To standardize the brightness of SNIa, each light curve is fitted with the SALT3 model from Kenworthy et al. (2021), the same model used in the simulations. This fitting process extracts the following parameters for each event: time of peak brightness (t_0), amplitude (x_0), stretch (x_1), color (c). For the low- z subset we used a fixed redshift z_{spec} . For the high- z subset we float z_{phot} as a fifth parameter using the host galaxy z_{phot} pdf as a prior, and using the fitting formalism in Kessler et al. (2010). SALT3 light curve fits for several events are shown by the smooth curves in Fig. 1 of M23.

Following Eq. 1 in Kessler et al. (2010), the 5-parameter SALT3 fit uses MINUIT (James and Roos 1975) to minimize the following χ^2 ,

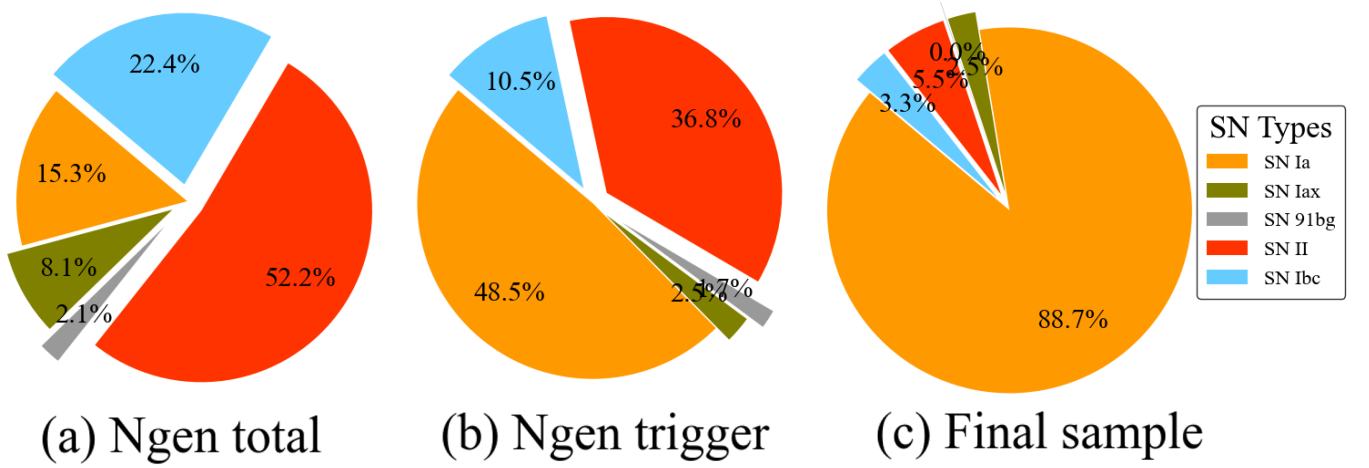


FIG. 2: Pie chart showing the percentage of true SN types. (a) Generated total, (b) Passed trigger and (c) After analysis cuts (these are the last three columns of Table III). The evolution of Ia fraction (orange) from left to right illustrates the increasing purity with detection and analysis.

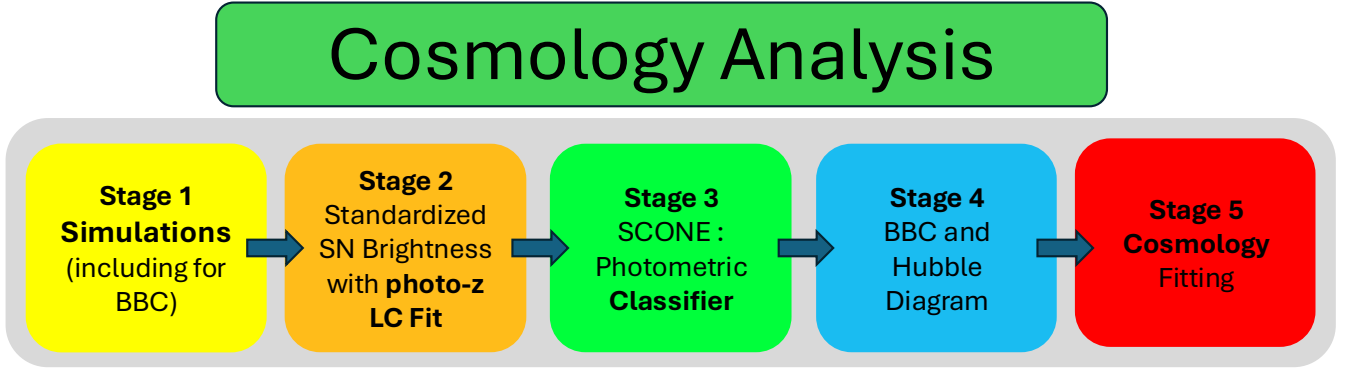


FIG. 3: Flowchart showing the cosmology analysis steps.

$$\chi^2 = \left(\sum_i \chi_{F_i}^2 \right) + \chi_{\text{host}}^2 + \chi_{\text{syst}}^2. \quad (4)$$

$\chi_{F_i}^2$ is the flux chi-squared for the i^{th} observation,

$$\chi_{F_i}^2 = \frac{[F_i^{\text{data}} - F_i^{\text{model}}(\vec{x}_5)]^2}{\sigma_{F,i}^2} + 2 \ln(\sigma_{F,i}/\tilde{\sigma}_{F,i}) \quad (5)$$

where F_i^{data} is the flux for the i^{th} observation and F_i^{model} is the SALT3 model flux computed from the fitted parameters $\vec{x}_5 = \{t_0, x_0, x_1, c, z_{\text{phot}}\}$. In the second term of eq. 5, $\sigma_{F,i}$ is the quadrature sum of the flux and model uncertainties, and $\tilde{\sigma}_{F,i} = \sigma_{F,i}$ after the first fit iteration so that the $2 \ln(\sigma_{F,i}/\tilde{\sigma}_{F,i})$ term is close to zero in the second and third fit iteration.

The contribution from the host galaxy prior is

$$\chi_{\text{host}}^2 = -2 \log[\mathcal{P}_{\text{host}}(z_{\text{phot}})] \quad (6)$$

where $\mathcal{P}_{\text{host}}(z_{\text{phot}})$ is the host galaxy probability for the redshift value to be z_{phot} .

The terms $\chi_{F_i}^2$ and χ_{host}^2 are the conventional terms defined in Kessler et al. (2010) and used in many z_{phot} studies including M23. Here we add a new term, χ_{syst}^2 to address a subtle artifact from fitting with systematics in Sec. IIIB. The underlying artifact is that a small systematic perturbation (e.g., changing filter zeropoint) can lead to an unphysical large change in SALT3-fitted values that result in either rejecting the event, or assigning such a large systematic uncertainty that the event is effectively ignored in a cosmology fit. While this artifact occurs rarely for any one systematic, the cumulative impact from many systematics can lead to a loss reaching $\sim 10\%$, and this loss is not currently modelled in our BBC method. To reduce this loss, we use the reference fitted parameters without a systematic shift as a prior in the SALT3 fit for each systematic shift,

$$\chi_{\text{syst}}^2 = [(p_k - p_k^{\text{ref}})/\sigma_k^{\text{ref}}]^2 \quad (7)$$

where p_k^{ref} are the fitted parameters ($k = 1, 5$) for the reference, p_k are for fits with a systematic perturbation, and σ_k^{ref} is the fitted uncertainty from the reference fit.

After each MINUIT fit iteration, the wavelength range for each LSST passband is transformed to the rest-frame using fitted z_{phot} . If the central rest-frame wavelength $\langle\lambda_{\text{cen}}\rangle$ is outside of the SALT3 model range (2800-17000Å), the passband is dropped in the next fit iteration; a previously dropped passband can be added if it is within the model wavelength range. If any passband is dropped or added, the fit iteration is repeated to ensure that a consistent set of passband is included in the fit.

B. Selection Requirements & Systematic Uncertainties

We apply the following selection requirements (cuts) based on previous analyses:

1. at least three bands with maximum signal to noise ratio $\text{SNR} > 4$
2. $|x_1| < 3.0$
3. $|c| < 0.3$
4. stretch uncertainty $\sigma_{x1} < 1.0$
5. time of peak brightness uncertainty $\sigma_{t0} < 2.0$ days
6. $P_{\text{fit}} > 0.05^7$. No explicit cuts are applied on SCONE classification.
7. $0.01 < z_{\text{phot}} < 1.4$
8. Fitted bands satisfy $2800 < \langle\lambda_{\text{cen}}\rangle/(1 + z_{\text{phot}}) < 17000$ the valid range of the rest frame wavelengths for the SALT3 model; same cut is applied for $z_{\text{phot}} \pm \sigma(z_{\text{phot}})$
9. valid bias correction for all systematic variation (see section §III D).

The statistics after cuts are presented in the last column of Table III, and the transient type fractions (for high- z) are shown in Fig. 2. The high- z true-Ia fraction is 15% for physical rates (Fig. 2a), increases to $\sim 50\%$ after two detection trigger (Fig. 2b), and increases again to $\sim 90\%$ after selection cuts (Fig. 2c).

The z_{phot} residual vs. z_{true} is shown in Fig. 4a for all galaxies in the catalogue, and in Fig. 4b for host galaxies after SN Ia trigger and selection cuts. Fig. 4c shows the performance for SN + Host combined fit. It is this sample that is used in the Hubble diagram. The core resolution is much better ($\sim 3\times$ better) than for the galaxies in Fig. 4a, although the outlier fractions are comparable. Figure 5 shows the redshift distribution in the final cosmology sample for 1 of the 25 samples, used for Hubble diagram.

The systematics considered in this analysis are identical to those used in Mitra et al. (2023). Table IV summarizes the systematics used and their corresponding values.

⁷ P_{fit} is the SALT3 fit probability computed from χ^2 and number of degrees of freedom.

TABLE IV: Source of Systematic Uncertainty

Row	Label	Description	Value ^a
1	StatOnly	no systematic shifts	—
2	MWEBV	shift $E(B - V)$	5%
3	CAL_HST	HST calibration offset	$0.007 \times \lambda$
4	CAL_ZP	LSST zero point shift	5 mmag
5	CAL_WAVE	LSST Filter shift	5 Å
6	zSPEC	shift z_{spec} redshifts	5×10^{-5}
7	zPHOT	shift z_{phot} redshifts	0.01
8	zPHOTERR	scale host z_{phot} uncertainty	1.2

^a Shift (or scale) applied to simulated data before each re-analysis

The systematics include variations in galactic extinction, calibration and host photo- z . The first 7 cuts are applied only to the nominal analysis without systematics; for each systematic variation we do not apply these 7 cuts but instead process the list of events passing cuts from the nominal analysis. For example, if an event has fitted SALT3 color parameter $c = 0.299$, and migrates to $c = 0.3001$ for a calibration systematic, this event is preserved without applying cuts that require $|c| < 0.3$.

Requiring valid fit bands (cut 8) for each systematic results in a 3.3% loss from requiring the same events passing all cuts. Requiring a valid biascor (cut 9 and Sec. III D 2) for each systematic results in a 0.51% loss.

1. Host Galaxy Mis-association

Host galaxy matching is done via the Directional Light Radius (DLR) method. A normalized, dimensionless parameter d_{DLR} is defined as the ratio of the angular separation $\Delta\theta$ between the supernova (SN) and DLR defined as the galaxy's centroid to the galaxy's effective radius in the direction of the SN. DLR is derived from the smeared Sersic profile. Mathematically, this is expressed as:

$$d_{\text{DLR}} = \frac{\Delta\theta}{\text{DLR}}.$$

For galaxies satisfying $d_{\text{DLR}} < 7$, the two galaxies with smallest d_{DLR} are used for analysis. A host match is defined as the galaxy with smallest d_{DLR} and $d_{\text{DLR}} < 4$.

Table V compares the host galaxy mismatch statistics of Host 1 with the numbers reported in Qu et al. (2024). Figure 6 also illustrates this statistics along with the distributions of DLR and r -band magnitude and host confusion parameter.

C. Photometric Classification: SCONE

We used SCONE (Supernova Classification with a Convolutional Neural Network) (Qu et al. 2021) as our photometric classifier for classifying between Ia and non-Ia SNe. SCONE is a neural network based classifier, which requires photometric data only, without the need

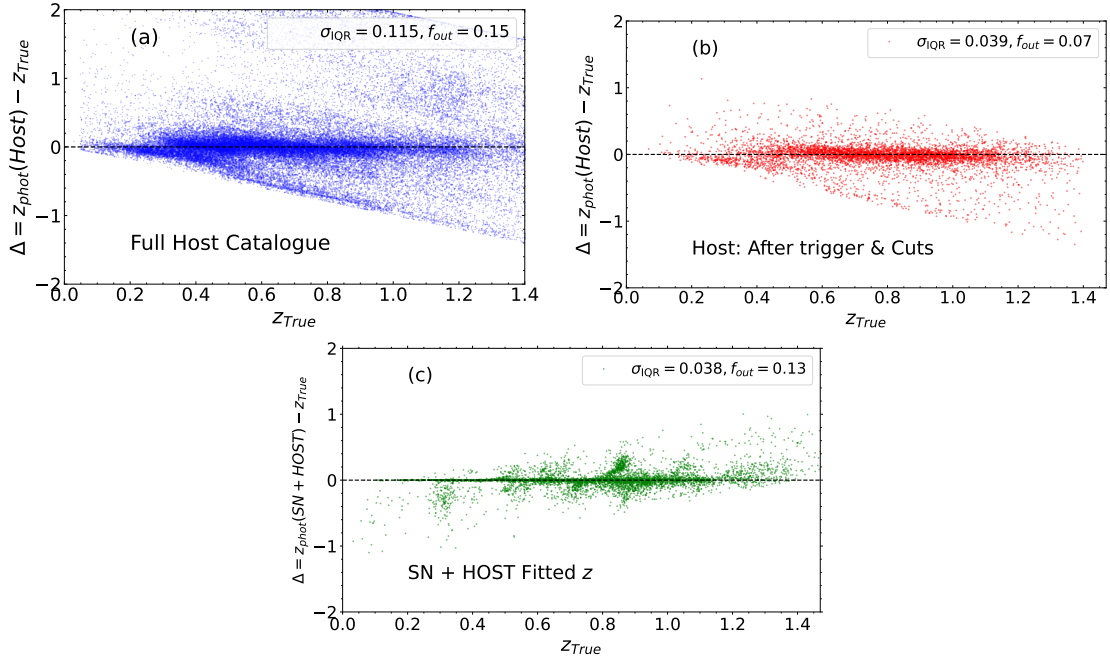


FIG. 4: Photo- z residual ($z_{\text{phot}} - z_{\text{true}}$) vs. z_{true} for (a) full host galaxy catalogue, (b) subset of host galaxy catalogue after trigger and cuts, (c) Combined SALT3 fitted SN + HOST photo- z . The σ_{IQR} and f_{out} numbers on each panel are computed for $0.4 < z_{\text{true}} < 1.4$. The source of photo- z is indicated on each vertical axis label.

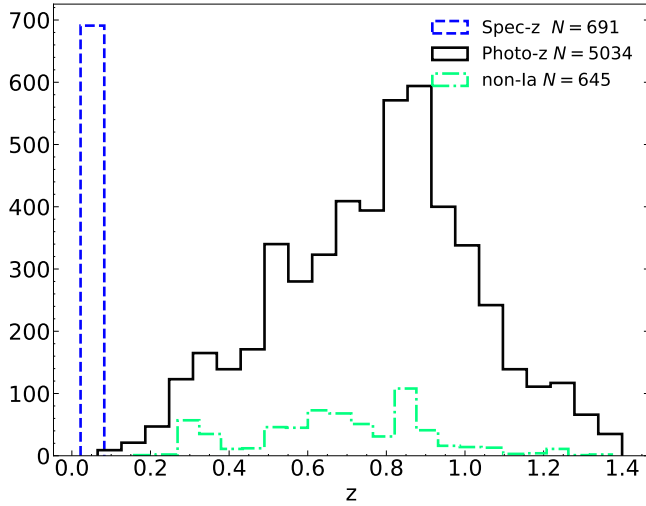


FIG. 5: Histogram of the z distribution of one of the twenty five SN sample Hubble diagram.

Condition	LSST SIM	LSST SIM II ^a	DES	Cosmology sample
No Host	6 %	2 %	2 %	No
Host	85.82	56.16	98	Yes
$N_{\text{match}}=2$ all ^b	11.96	7.93	9	Yes
$N_{\text{match}}=2$ wrong ^c	1.04	0.75	—	Yes
Host depth	25.4		25.5	NA

^a Compared to DES $z \leq 0.8$ range.

^b host 2 $d_{\text{DLR}} < 4$

^c The galaxy with the smaller d_{DLR} is the incorrect host.

TABLE V: Statistics of host galaxy (mis)match (in %) in this analysis (first column) and compared with DES analysis from Qu et al. (2024) (second column). The third column tells us whether the corresponding row is used in the cosmology analysis. The value marked as ‘—’ signifies that we could not get this number from Qu (2024).

for any accurate redshift approximation. Deep learning based photometric classifiers have outperformed its predecessors such as light curve template matching algorithms (Sako et al. 2008) or feature extraction with either sequential cuts (Campbell et al. 2013) or machine learning methods (Möller et al. 2016, Lochner et al. 2016, Dai et al. 2018). SCONE has relatively low computational and dataset size requirements for achieving high accu-

racy.

We use SCONE as a binary classifier to determine the probability, P_{Ia} , that each event is a type Ia SN. For training the network with 150 iterations, we generate a simulation consisting of $\sim 50,000$ Ia and $\sim 50,000$ non-Ia after selection requirements. Even for real data, the procedure of training on simulated data is the same to ensure that the training data has the same redshift range and selection effects as the data. The training process took ~ 3 hours. The network attained 98% accuracy for both training and validation (Fig. 7). Fig. 8 shows P_{Ia} distri-

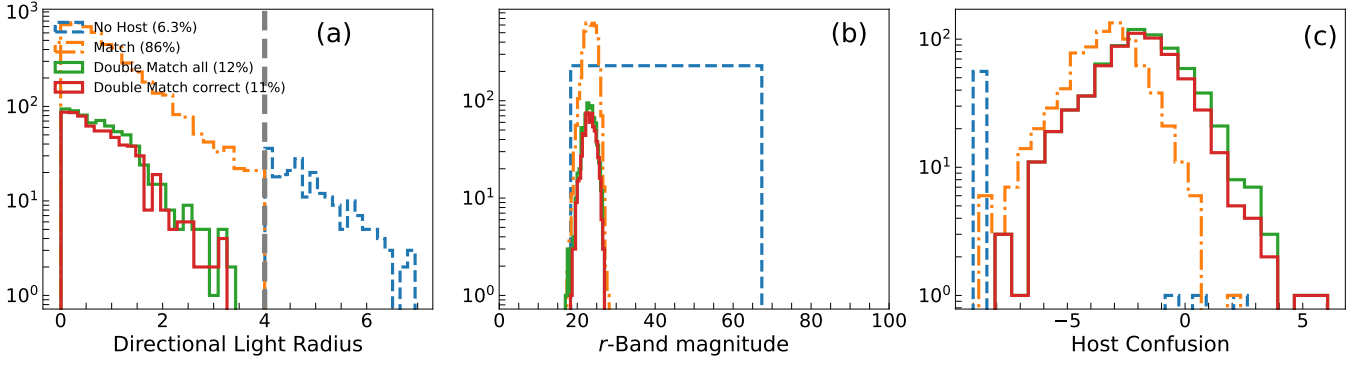


FIG. 6: Different statistics for host galaxy (mis)-association computation. We find (see Table V for a comparison with DES values from Qu (2024)) $\sim 86\%$ host galaxy match and $\sim 6\%$ of no host matches. 12% ‘double match all’, is the aggregate for all matches for $\text{HOST_d}_{\text{DLR}} < 4$ and 11% for the sum of all correct host matches, satisfying $\text{HOST_d}_{\text{DLR}} < 4$ condition. The grey vertical dashed line in the first figure marks $d_{\text{DLR}} = 4$.

butions and they match our qualitative expectations; for true Ia’s the probability is peaked at $p \sim 1$ and for true non-Ia the peak is at $p \sim 0$.

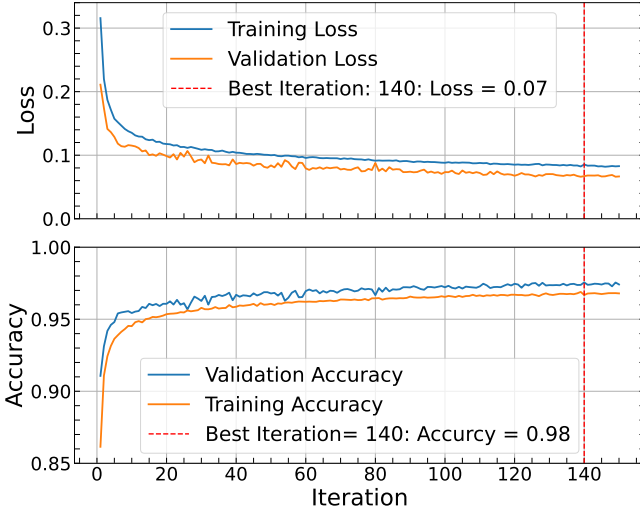


FIG. 7: Accuracy and loss score in the training phase of SCONE.

D. BEAMS with Bias Corrections (BBC)

BBC, an extension of BEAMS, produces a Hubble diagram that is corrected for selection effects and also corrected for non-Ia contamination. Using a photometrically classified sample, BBC was first used to obtain constraints for the DES-SN5YR cosmology analysis.

1. BEAMS

The BEAMS framework was introduced by Kunz et al.

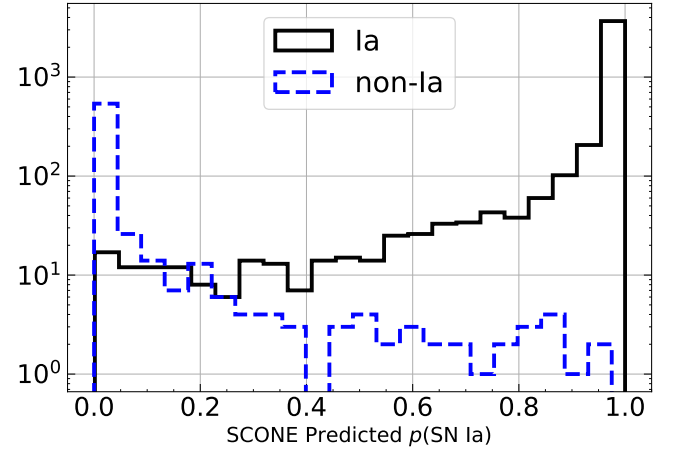


FIG. 8: Distribution of SCONE predicted probability of a SN candidate to be type Ia, for true SNIa (black histogram) and for true non-Ia (dashed blue histogram).

(2007), Newling et al. (2012), Hlozek et al. (2012). In the BEAMS framework, the total likelihood function includes two terms: one for the SN Ia population, $\mathcal{L}_{\text{Ia}}$, and another for the contaminant population, $\mathcal{L}_{\text{CC}}$ ⁸, expressed as:

$$\prod_{i=1}^{N_{\text{SNe}}} (\mathcal{L}_{\text{Ia}}^i + \mathcal{L}_{\text{CC}}^i). \quad (8)$$

The likelihood components for each event i are:

$$\begin{aligned} \mathcal{L}_{\text{Ia}}^i &= P_{\text{Ia}}^i \times D_{\text{Ia}}(z_i, \mu_{\text{obs},i}, \mu_{\text{ref}}), \\ \mathcal{L}_{\text{CC}}^i &= (1 - P_{\text{Ia}}^i) \times D_{\text{CC}}(z_i, \mu_{\text{obs},i}, \mu_{\text{ref}}), \end{aligned} \quad (9)$$

where D_{Ia} and D_{CC} are the SNIa and non-Ia Hubble residual likelihoods.

⁸ The CC notation includes all non-Ia, but here we follow the KS17 notation.

The quantities in Eq. 9 can be used to determine a BEAMS probability:

$$\mathcal{P}_{\text{BEAMS}} = \frac{P_{\text{Ia}}^i D_{\text{Ia}}^i}{P_{\text{Ia}}^i D_{\text{Ia}}^i + (1 - P_{\text{Ia}}^i) D_{\text{CC}}^i}, \quad (10)$$

which quantifies the probability of a given SN belonging to the Ia population versus the contaminant population, taking into account both the classification probability (P_{Ia}^i) and the Hubble residual information.

2. BBC

BBC is a fitting framework that accounts for selection effects using a simulation, incorporates BEAMS to account for non-Ia contamination, and implements the Marriner et al. (2011) method of measuring distances independently of cosmological parameters. BBC reads the SALT3 fitted parameters (high- z and low- z) from the data and biasCor simulation (sec. IIB), and produces a bias-corrected Hubble diagram, both unbinned and in redshift bins. For each event, the measured distance modulus is based on the Tripp (1998) equation,

$$\mu = m_B + \alpha x_1 - \beta c + \gamma G_{\text{host}} + M_0 + \Delta\mu_{\text{bias}}, \quad (11)$$

where $m_B \equiv -2.5 \log_{10}(x_0)$, $\{\alpha, \beta, M_0\}$ are global nuisance parameters, and $\Delta\mu_{\text{bias}} = \mu - \mu_{\text{true}}$ is determined from the biasCor simulation in a 4-dimensional space of $\{z, x_1, c, x_M\}$. A valid bias correction is required for each event, resulting in a $<1\%$ loss, usually for extreme values of x_1, c, z where the biascor statistics are very low. The term γG_{host} captures residual dependency between SNIa luminosity and host galaxy properties. The distance uncertainty (σ_μ) is computed from Eq. 3 of KS17.

There are two subtle issues concerning the use of z_{phot} and its uncertainty σ_z . First, the calculated distance error from σ_z (σ_μ^z in Eq. 3 of KS17) is an overestimate because it does not account for the correlated color error that reduces the distance error. By floating z_{phot} in the SALT3 fit, redshift correlations propagate to the other SALT3 parameter uncertainties, and therefore we set $\sigma_\mu^z = 0$. The second issue concerns the μ_{bias} computation, where μ_{true} is computed at SALT3-fitted z_{phot} rather than the true redshift.

To avoid a dependence on cosmological parameters, the BBC fit is performed in 14 logarithmically-spaced redshift bins. The fitted parameters include the global nuisance parameters (α, β, M_0) and bias-corrected distances in 14 redshift bins. The unbinned Hubble diagram is obtained from Eq. 11 using the fitted parameters, and the distance uncertainty is computed from Eq. (10) in Kessler et al. (2023).

3. Covariance Matrix

The uncertainties in SN distance measurements are en-

capsulated in a covariance matrix, which quantifies both statistical and systematic uncertainties. This covariance matrix ($\mathbf{C}$) is

$$\mathbf{C} = \mathbf{C}_{\text{stat}} + \mathbf{C}_{\text{syst}} \quad (12)$$

where $\mathbf{C}_{\text{stat}}$ is the diagonal covariance representing measurement errors. $\mathbf{C}_{\text{syst}}$ is the systematic covariance matrix (Conley et al. 2011),

$$C_{\text{syst}}^{ij} = \sum_S (\Delta\mu_{\text{obs},S}^i) (\Delta\mu_{\text{obs},S}^j), \quad (13)$$

where i and j are SN indices, and $\Delta\mu_{\text{obs},S}$ is the change in SN distance moduli resulting from systematic variation S .

Because of cuts 8 and 9 in Sec. IIIB, the SN sample for each systematic is slightly different and therefore $\mathbf{C}_{\text{syst}}$ cannot be computed. Following previous analyses we select the common events in all systematics and use this sample for the Hubble diagram and $\mathbf{C}$. For redshift systematics that result in migration to another redshift bin, the original (no syst) redshift bin is preserved for the BBC fit.

This implicit common-event cut is not modelled in the bias correction simulation because we do not perform the CPU-intensive task of evaluating systematics for the large biascor sample. This implicit cut rejects $\sim 4\%$ of the sample (2nd column of Table VI) and could be a potential source of unmodelled bias. As discussed in Sec. VA this choice of prior results in the smallest unmodelled loss of events.

TABLE VI: Analysis loss from requiring valid biasCor and requiring that the same events pass cuts for all systematic variations.

Data cuts not applied to biasCor sample	% Loss 1 σ prior ^a	% Loss 2 σ prior	% Loss Flat prior ^b
Analysis cuts on systematics	3.3%	3.9%	7.5%
Valid biascor in Nominal	0.44%	0.89%	0.86%
Valid biascor in systematics ^c	0.07%	0.37%	4.00%
Total	$\sim 4\%$	$\sim 5.2\%$	$\sim 12.4\%$

^a Prior refers to χ_{syst}^2 in Eq. 7.

^b All previous analyses using SNANA, including M23 and DES-SN5YR, used a flat prior.

^c for stat-only analysis without systematics, this loss is zero by definition.

4. Cosmology Fitting and Figure of Merit

In cosmology fitting, we minimize the following χ^2 :

$$\chi^2 = \Delta\mu^T \mathbf{C}^{-1} \Delta\mu, \quad (14)$$

where C is the covariance matrix (Eq. 12), and $\Delta\mu$ is the vector of differences between observed and theoretical distance moduli. We use the same fast minimization code⁹ that was used in Mitra et al. (2023). We approximate a CMB prior using the R -shift parameter (e.g., see Eq. 69 in Komatsu et al. (2009)) computed from the same cosmological parameters that were used to generate the SNe Ia. The R -uncertainty is $\sigma_R = 0.006$, tuned to approximate the constraining power of Planck Collaboration et al. (2020). We fit with w CDM and w_0w_a CDM (CPL (Chevallier and Polarski 2001)) models, where $w = [w_0 + w_a(1 - a)]$ and $a = 1/(1 + z)$.

While Brout et al. (2021) showed that fitting an unbinned Hubble diagram gives optimal constraints, the CPU-time increases significantly for large Hubble diagrams. Kessler et al. (2023) suggested a rebinned approach to optimise constraints with a much smaller Hubble diagram, and hence use much less CPU resources. We include rebinned results here with 4 stretch bins, and 8 colour bins. The maximum Hubble diagram size is $14 \times 4 \times 8 = 448$; after excluding empty bins the average fitted Hubble diagram size is 300.

For the w_0w_a CDM model, the FoM is computed based on the dark energy task force (DETF) definition in Albrecht et al. (2006),

$$\text{FoM} \simeq \frac{1}{\sigma(w_a)\sigma(w_0)\sqrt{1 - \rho^2}}, \quad (15)$$

where ρ is the reduced covariance between w_0 and w_a .

We fit the Hubble diagram and evaluate FoM for the three binning methods: binned, unbinned, and rebinned. M23 reported that unbinned results were pathological for unknown reasons and therefore not shown. Here we find sensible unbinned results but we cannot trace what caused the problem in M23.

IV. COSMOLOGY RESULTS

For one of the 25 statistically independent samples, we show the Hubble diagram produced by the BBC fit, both binned and unbinned, in Fig. 9. The Hubble residuals with respect to the true cosmology, $\Delta\mu = \mu - \mu_{\text{true}}$, are consistent with zero and do not show a redshift-dependent slope.

For BBC-fitted nuisance parameters we find $\alpha - \alpha_{\text{true}} = -0.00095 \pm 0.00057$ (averaged over 25 samples), we cannot however trace the bias for β since β_{true} is unknown as described previously in Sec. II B. We find $\sigma_{\text{int}} = 0.112$ but there is no true σ_{int} for comparison. Next, we show the BBC fitted distance uncertainties (σ_μ) in redshift bins (Fig. 10b). Owing to the larger statistics in high- z the distance uncertainty is in 0.01 – 0.02 range at $z > 0.6$.

For the subsections below, we define w -bias to be $w - w_{\text{true}}$ where w is from the w CDM cosmology fit. A similar definition is used for w_0 and w_a for the w_0w_a CDM model. The bias uncertainty is the standard deviation of the w -bias values divided by $\sqrt{25}$.

A. w CDM Results

For the w CDM cosmology fits, Table VII shows the average w -bias and average uncertainty among the 25 samples with a CMB prior. The w -bias precision is ~ 0.004 . The average w -uncertainty ($\langle\sigma_w\rangle$) for the whole sample is 0.025 (stat + syst), compared to $\langle\sigma_w\rangle = 0.020$ for the stat only case. Statistical uncertainties appear to dominate the systematic case. The addition of CMB prior on SN measurements alleviates the impact of systematics. This is primarily due to the orthogonality of direction of these two probe's measurements on the $\Omega_m - w$ plane for w CDM cosmology. Fig. 11 shows the w bias for all the three binning schemes.

TABLE VII: Summary of w CDM Cosmology bias for Binned, Unbinned, and Rebinned Cases

Case	Stat only		Stat+Syst	
	$\langle w\text{-bias} \rangle^a$	$\langle \sigma_w \rangle^b$	$\langle w\text{-bias} \rangle$	$\langle \sigma_w \rangle$
Binned	0.0120 ± 0.0047	0.0205	0.0048 ± 0.0046	0.0251
Unbinned	0.0183 ± 0.0032	0.0202	0.0140 ± 0.0035	0.0217
Rebinned	0.0184 ± 0.0049	0.0201	0.0119 ± 0.0051	0.0192

^a Average bias among 25 samples with uncertainty of $\text{std}/\sqrt{25}$

^b Average fitted uncertainty among 25 samples.

B. w_0w_a CDM Results

For the w_0w_a CDM model, the average w_0 , w_a -bias, uncertainty, and FoM are shown in Table VIII for all three binning methods. For stat only all three binning methods have a similar $\sim 2\sigma$ bias (crosses in Fig. 12a), and the $\langle\text{FoM}\rangle$ values agree to within 1%. While the consistency among binning methods matches our expectation, the evidence of a small bias suggests a subtle issue with the analysis.

For stat+syst, binned and rebinned result in few σ biases on w_0 , w_a , and the unbinned bias is significantly larger. While these biases are significant because of the large number of datasets, the magnitude of the bias is significantly smaller than the average uncertainty for a single sample shown by the blue contours in Fig. 12a (stat only) and Fig. 12b (stat+syst).

For the binned analysis, stat+syst $\langle\text{FoM}\rangle = 126$, a 30% degradation compared to the stat-only $\langle\text{FoM}\rangle$. The rebinned analysis improves $\langle\text{FoM}\rangle$ to 153, and the unbinned analysis gives the optimal $\langle\text{FoM}\rangle = 160$. The rebinned analysis has a significant $\langle\text{FoM}\rangle$ improvement over the

⁹ <https://github.com/RickKessler/SNANA/blob/master/src/wfit.c>

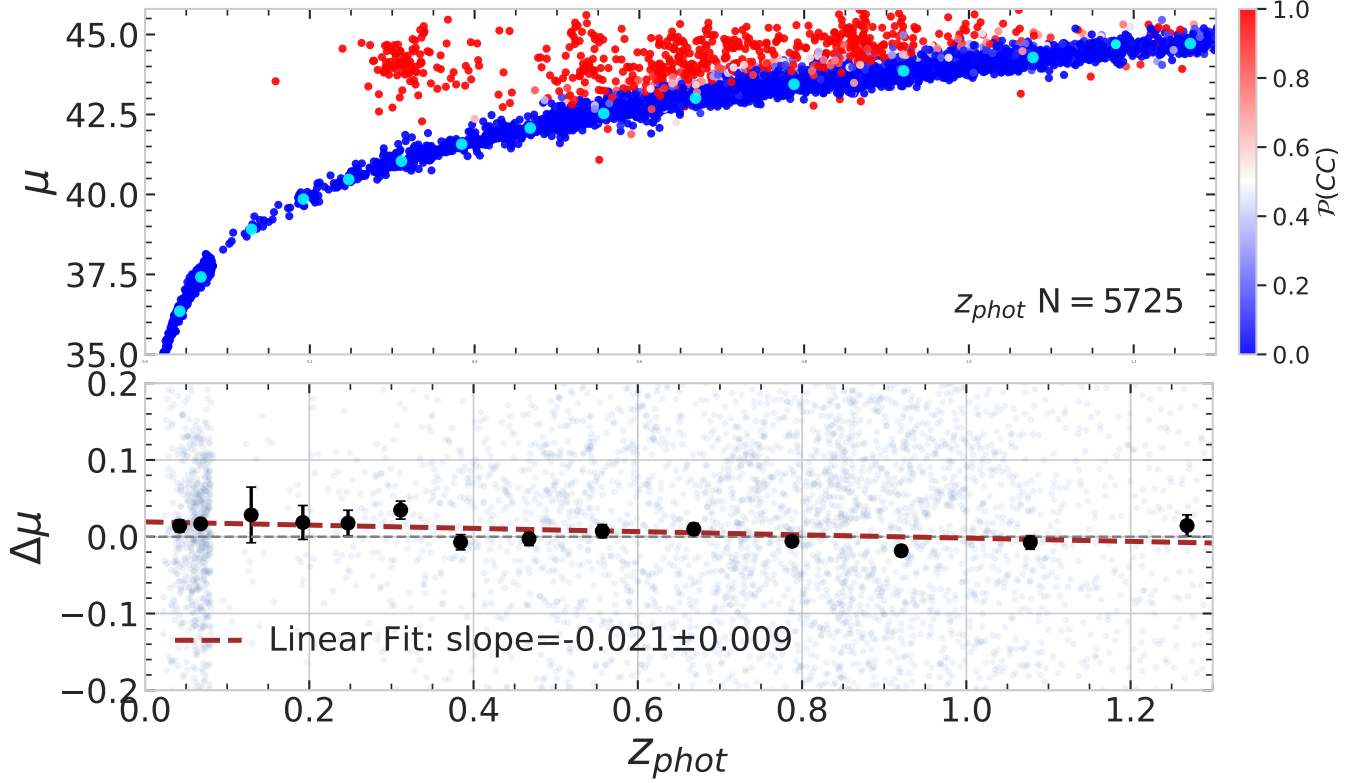


FIG. 9: Redshift binned and unbinned Hubble diagram for one of the 25 statistically independent samples. The lower panel shows Hubble residual $\Delta\mu$ with respect the true cosmology: the error bar shows the rms in each BBC redshift bin (same redshift bins as in Fig. 10). The observed red-coloured events, show the contaminations from the core-collapse SNIa. The corresponding colour bar on the right shows the probability derived from BEAMS. The number of events with $\mathcal{P}(\text{CC}) > 0.75$ is 652, which is $\sim 11.4\%$ of the whole z_{phot} data sample of 5725 events.

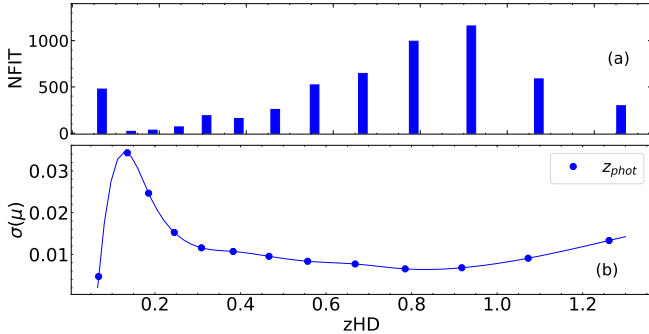


FIG. 10: Number of events (top) and BBC-fitted distance uncertainty (bottom) per redshift bin.

$$\mathcal{R}_{\text{FoM},i} = \text{FoM}_{\text{syst},i} / \text{FoM}_{\text{stat}}, \quad (16)$$

where $\text{FoM}_{\text{syst},i}$ is the FoM from including only systematic i , and FoM_{stat} is the FoM without systematic uncertainties. Note that $\mathcal{R}_{\text{FoM},i} \leq 1$.

Table IX compares the FoM degradation while moving from stat only to stat+syst case for the binned HD. It is seen that the FoM degradation is dominated by the calibration systematics. Photo- z systematics are not the dominant source of FoM degradation. Primarily because the low- z region is most sensitive to redshift errors, is mainly populated by spectroscopic redshifts, which are unaffected by this systematic. Additionally, an observed anti-correlation between photo- z and the SALT3 color parameter acts to self-correct distance errors. Similar findings were reported by [Chen et al. \(2022\)](#). Thus, the overall impact of photo- z systematics remains modest due to these combined effects.

C. Comparison with DES-SN5YR and DESI

In Fig. 13, we provide constraints on dark energy equation of state w_0 and its redshift evolution w_a excepted

binned analysis, but further improvement is still possible as indicated by the unbinned result.

Since there are many systematics contributing to the decrease in $\langle \text{FoM} \rangle$, we quantify the impact of each systematic “ i ” by recomputing the covariance matrix separately for each systematic ($\mathbf{C}_{\text{syst},i}$), and repeating the cosmology fit for each $\mathbf{C}_{\text{syst},i}$. We finally compute the FoM ratios

TABLE VIII: Summary of w_0w_a CDM Cosmology Fits: Binned, Unbinned, and Rebinned Analysis.

	Stat only			Stat+Syst		
	Binned	Unbinned	Rebinned	Binned	Unbinned	Rebinned
$\langle w_0\text{-bias} \rangle^a$	-0.0226 ± 0.0150	-0.0156 ± 0.0144	-0.0297 ± 0.0145	-0.0256 ± 0.0148	-0.0531 ± 0.0151	-0.0277 ± 0.0136
$\langle w_a\text{-bias} \rangle$	0.1423 ± 0.0629	0.1399 ± 0.0609	0.2004 ± 0.0628	0.1198 ± 0.0642	0.2800 ± 0.0617	0.1594 ± 0.0596
$\langle \sigma_{w_0} \rangle^b$	0.0640	0.0638	0.0647	0.0776	0.0730	0.0729
$\langle \sigma_{w_a} \rangle$	0.2736	0.2718	0.2730	0.3242	0.2986	0.3053
(FoM)	183	187	188	126	160	153

^a Average bias among 25 samples with uncertainty of $\text{std}/\sqrt{25}$

^b Average fitted uncertainty among 25 samples.

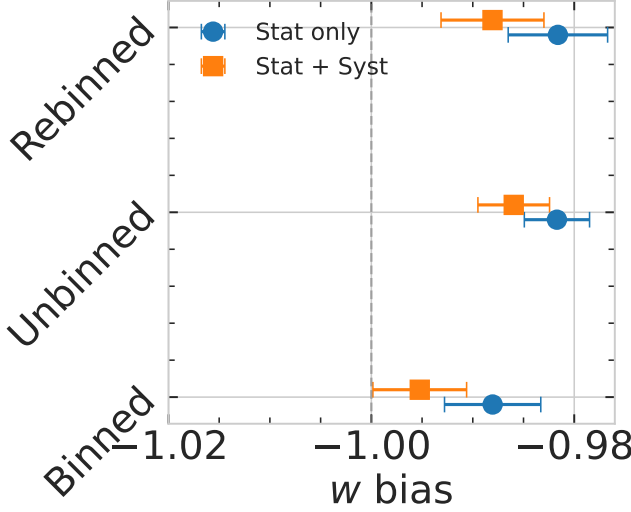


FIG. 11: Bias in w CDM cosmology. The grey dashed vertical line gives the reference $w = -1$. In DES-SN5YR, Vincenzi et al. (2024) reported w bias as 0.017.

TABLE IX: FoM-Ratio $\mathcal{R}_{\text{FoM},i}$ for Each Systematic (w_0w_a CDM model) in binned, unbinned, and rebinned cases.

Systematic(s)	$\mathcal{R}_{\text{FoM},i}$		
	Binned	Unbinned	Rebinned
None (stat only)	1.00	1.00	1.00
zSHIFT	0.94	0.99	0.98
ZERRSCALE	0.92	0.99	0.98
Photo_shift	0.92	0.99	0.98
MWEBV	0.93	0.99	0.98
CAL_WAVE	0.84	0.97	0.95
Cal_ZP	0.78	0.91	0.89
CAL	0.72	0.87	0.83
Stat + All Syst	0.70	0.86	0.81

from the binned LSST SNe sample from this work when combined with DESI-DR2 baryonic acoustic oscillations (BAO) measurements (DESI Collaboration et al. 2025) and *Planck* PR4 CMB power spectrum measurements

(CamSpec) likelihood, Rosenberg et al. 2022). To be consistent with the LSST SNe simulations, we replace the actual DESI BAO data with the $w_0w_a\Lambda$ CDM expectations with values set to *Planck* 2018 cosmology and $(w_0, w_a) = (-1, 0)$. The measurement covariance is same as the ones released by DESI DR2. More details about this can be found in Raghunathan and Mitra (2025). As a comparison, we also show the actual constraint obtained by DES-SN5YR SNe when combined with DESI-DR2 BAO in red, *Planck*-CMB in blue, and the joint constraint from all three in green. When replacing the DES-SN5YR SNe sample with LSST, we expect the figure-of-merit on w_0, w_a parameters to improve by $\times 16$ for SNe-only and by $\times 1.4$ for the combination with BAO and CMB.

V. DISCUSSION OF BIASES

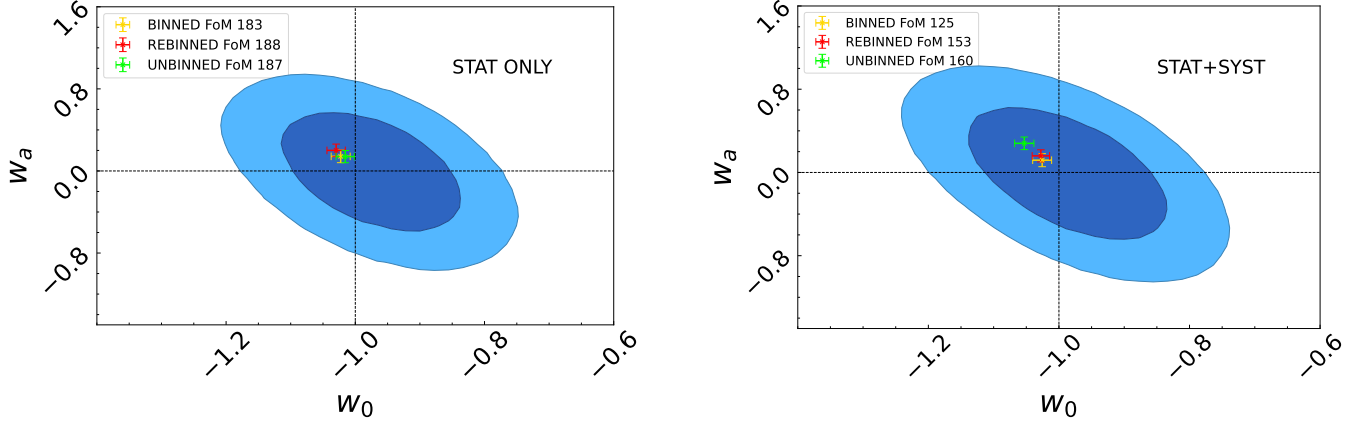
A. Robustness Tests

In previous cosmology analyses using SNANA and pipin, a flat prior was used to refit (with SALT3) each sample with systematic variants. They did not report the corresponding loss that is not accounted for in the bias corrections, but here we have investigated this loss and find it to be 12% (last column in Table VI). We therefore developed and used the χ^2_{syst} prior in 7, which reduces this loss to 4%, and we also tested using $2\sigma_k^{\text{ref}}$ in Eq. 7. and also tested a 2σ prior resulting in a 5% loss. NEED INTERNAL TABLE OF W0,WA BIAS VS. PRIOR.

VI. CONCLUSIONS

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Author contributions are listed below.
A. Mitra: co-lead project, SNANA simulations and analysis, writing
R. Kessler: co-lead project, software, analysis, writing
H. Qu: SCONE software support
P. Armstrong: Pippin software support
S. Raghunathan: Analysis, plotting



(a) w_0 - w_a 2σ (95% confidence) contours for a single SNIa data sample combined with CMB prior. Contours for unbinned HDs are shown. Shown for stat only case. The crosses show the $\pm 1\sigma$ bias from averaging results over the 25 simulated data samples.

(b) Same as in let figure but for stat+syst case.

FIG. 12: Comparison of 2σ (95% confidence) contours for two cases: (a) stat only, and (b) stat+syst.

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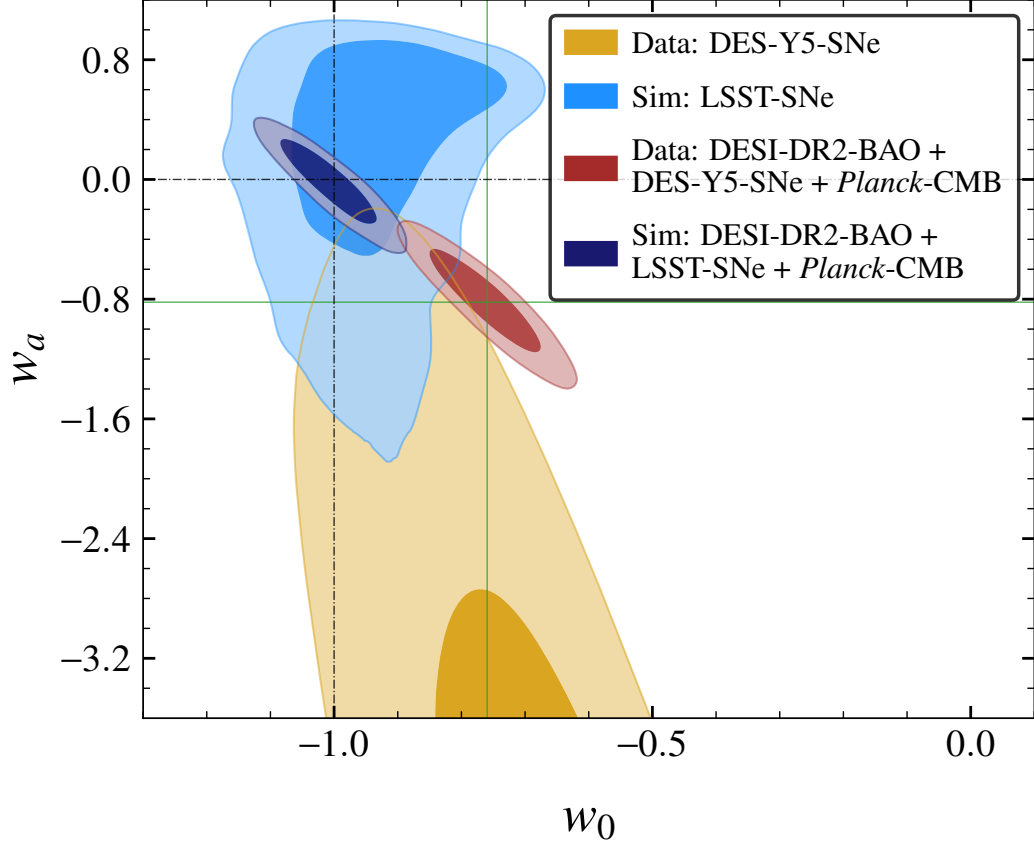


FIG. 13: Constraints on dark energy equation of state w_0 and its evolution w_a when combining different datasets: BAO data from DESI-DR2 release in orange; its combination with either DES-SN5YR (*Planck*-CMB) in red (blue); and the joint constraints from all the three datasets in green. The best-fit values $(w_0, w_a) = (-0.76, -0.82)$ for the joint combination are marked with green lines for reference. We also show the constraining power expected when combining with a binned HD sample from the current analysis, with mock DESI-DR2-like BAO and *Planck*-CMB in black. As it is evident from the figure, replacing DES-SN5YR with LSST Hubble diagram, can reject the deviation from standard Λ CDM, as inferred by green, with $\geq 5\sigma$ confidence.

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